

R Sai Pranav

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SUMMARY

CS student building **Generative AI** and **Machine Learning** systems for healthcare and documents: fine-tuned an open-weight LLM (Qwen2.5-3B) with **LoRA/PEFT** on Hugging Face, built **RAG** pipelines on **vector databases** (FAISS), designed **agentic LLM workflows** with **LangChain**, **LangGraph**, Jev / Laya decision models, and **prompt engineering** over the **OpenAI** and **Anthropic (Claude)** APIs, and trained deep learning models from scratch in **PyTorch**, including an explainable-AI breast-cancer pipeline (paper in progress).

EXPERIENCE

SDE Intern, MES — Cepheid India Pvt Ltd (Danaher Corporation), Bangalore/Gurugram *Sep 2026 – Present*

- Develop and maintain production **Web APIs** (REST) integrating with **Siemens Opcenter MES**, customized to business requirements, working with structured manufacturing data in **SQL Server** on **AWS** infrastructure.
- Work in an **Agile** model with monthly sprints across the full software development lifecycle, from user stories to iterative deployment; maintain the reliability of code running in live manufacturing, collaborating with business and engineering teams.

Research Intern, Explainable AI — CSIR-4Pi (Fourth Paradigm Institute), Bangalore *Jul 2025 – Jan 2026*

- Trained and evaluated **deep learning** models from scratch in **PyTorch** (Attention U-Net, ResNet-U-Net, U-Net) for breast-cancer segmentation and detection on **healthcare** medical imaging (MedSAM, ResNet backbones).
- Diagnosed model failures and monitored performance to improve reliability; validated on noisy, unstructured real-world data, using **synthetic data generation** / augmentation to address limited labelled data.
- Built **explainable-AI (XAI)** heatmaps so clinicians could interpret model predictions; co-authoring a research paper on this work.

Software Engineering Intern, Full Stack — MyStartupWave, Bangalore *Dec 2025 – Present*

- Built and shipped 10+ production applications (React, **FastAPI**, Node.js, MongoDB); cut API response times up to 35% via caching and bottleneck fixes; wrote unit/integration tests and automated API tests (Postman).

Project Intern, Cybersecurity Engineering — iSPIRT Foundation (Sakram), Bangalore *Nov 2025 – Present*

- Building protocol frameworks for India's Digital Public Infrastructure (DPI) in C++ and Python; contributing to hardware-bound cryptographic validation and data-integrity verification.

PROJECTS & ACHIEVEMENTS

Mortgage Document Intelligence Pipeline — InfrX 2026 (Infrdd), 2nd Place (₹75,000)

- Built a 5-stage **Generative AI** pipeline segmenting unstructured 100–2,000-page mortgage packages into 27 document types at **91% accuracy** (1,845 pages), using an **agentic** cost-aware routing cascade (pdfplumber → heuristics → **GPT-4o-mini** VLM via **OpenAI API** with **prompt engineering**) so most pages never reach the LLM.
- Designed Probabilistic Table Threading, a **Naive Bayes** statistical model fusing 5 signals to stitch table fragments split across pages; auto-merges ~85% of pairs, escalating ambiguous cases to an **LLM** arbiter. Dual-mode OCR (VLM + Tesseract / zero-cost Tesseract-only), validated on 2,295 pages.

CogniSure — Insurance Document Analyzer · HackRx 6.0 (Bajaj)

- Built a **Retrieval-Augmented Generation (RAG)** system for insurance-policy Q&A: **NLP** semantic clause retrieval with sentence-transformers (all-MiniLM-L6-v2) embeddings and a **FAISS vector database** behind a Flask API for coverage reasoning.

LLM Fine-Tuning (LoRA / PEFT) — Personal Project

- Fine-tuned an open-weight **Large Language Model** (Qwen2.5-3B) with **LoRA / PEFT** and Hugging Face Transformers on a custom instruction dataset built from medical-segmentation (MedSAM) work.

OrbitClean — AWI SpaceTech Hackathon, Top 5

- Prototyped a **Scikit-Learn** Random Forest classifier on Sentinel-2A satellite imagery to detect illegal dumps, reaching **AUC 0.80** over a 552-cell grid; built a Clarke-Wright route-optimization engine.

Decentralized Spatial-Temporal Data Veracity System — Personal Patent, Filed 2026

- Verifies spatial-temporal data in distributed ledgers via geospatial filtering, node credibility scoring, and cryptographic checks.

TECHNICAL SKILLS

Languages: Python, C++, Java, SQL, HTML/CSS

Generative AI / LLMs: LLMs, Prompt Engineering, Agentic AI Workflows, RAG Architectures, Vector Databases (FAISS), LangChain, LangGraph, Jev, Laya, OpenAI API, Anthropic (Claude) API, LLM Fine-Tuning (LoRA / PEFT), Hugging Face, VLMs

ML / Deep Learning: PyTorch, TensorFlow, Scikit-Learn, Machine Learning, Deep Learning, CNNs, U-Net / Attention U-Net, NLP, Computer Vision, OCR, Document Understanding, Explainable AI, Synthetic Data Generation, Model Training & Evaluation

Data Science: Pandas, NumPy, Statistics, Data Analysis, Structured & Unstructured Data Processing, Problem Solving, Rapid Prototyping

Tools, Cloud & MLOps: MLOps, FastAPI, Flask, SQL Server, MongoDB, Docker, Git, AWS, Google Cloud, Siemens Opcenter MES, REST APIs, Microservices, CI/CD, Agile / Scrum, DSA

EDUCATION

REVA University, Bangalore — B.Tech, Computer Science Engineering · GPA: 8.54 / 10.0 *2023 – 2027*

LEADERSHIP & INVOLVEMENT

Technical Lead, Algorand Club, REVA: organised 5+ hackathons, competed in 16+ (Google DeepMind, Adobe, HackRx); GDSC/GDG, AWS Cloud Club.